GRAPHITE STRAIN SENSOR

Technical datasheet

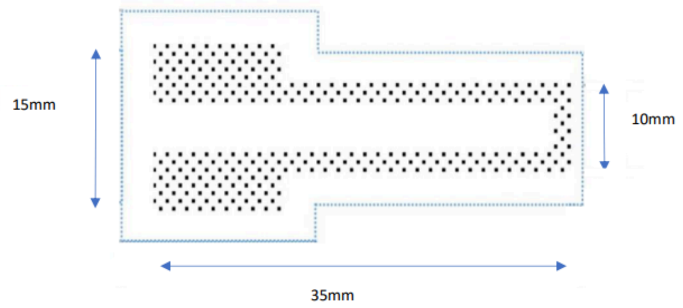

Figure 1: Dimensions of the sensor

Figure 2: Picture of the sensor connected

1. Main features

- Low cost
- Low tech
- Open source
- Homemade
- Transportable
- Light weight

2. General description:

Developed at INSA's Physics Department, this strain sensor consists of a paper substrate with a graphite layer deposited with a pencil. It works due to the quantum tunneling effect between these graphite particles: mechanical deformation (tensile or compressive strain) alters the inter-particle distance, which changes the tunneling probability and induces a measurable change in the device's electrical resistivity, (Lin et al., 2014) known as piezoresistive effect (Figure 1).

The variation of resistivity of the graphite when deformed is used to make a strain sensor.

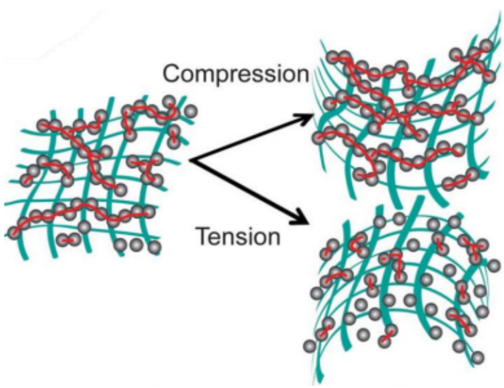

Figure 3: Particle distribution when tension and compression deformations are applied (Lin et al., 2014)

3. Specifications

Type	*Strain sensor*	Measurand	*Resistance*
Materials	*Graphite, paper*	Output signal	*Analog*
Type of sensor	*Passive*	Power supply	*5V*
Graphite compatibility	*HB, B, 3B*	Response time	*> 100ms*

Tab 1: Practical specifications of the strain sensor

4. Electrical characteristics

In tab 2 we can find typical values for resistivity measured for different pencil hardnesses without any deformations applied. As the amount of deposited graphite can't be really measured, these values are very hard to repeat (resistivity is very sensitive to the amount of graphite we deposit on our sensor (Lin et al., 2014). For example, when testing B characteristics, we deposited more graphite between tension and compression tests, which changed our flat resistance from 16500 Ω to 39000 Ω (Tab 2).
On the other hand, if we don't deposit more graphite, flat resistivity tends to decrease, as shown in 3B example. After several measures taken with the same amount of graphite, the flat resistance changed from 220kΩ to 71kΩ. In addition, the values depend on our amplification system (figure 4), and therefore on our potentiometer value, which has been changed for different hardnesses for having more precise values (R pot).

			Compression			**Tension**		
	R (pot)	Units	Min	Typ	Max	Min	Typ	Max
Voltage	-	V	-	5	-	-	5	-
Resistivity HB	5052 Ω	kΩ	58	64	70	82	86	91
Resistivity B	1028 Ω	kΩ	13	16,5	17	33	39	41
Resistivity 3B	10012 Ω	kΩ	210	220	224	65	71	73

Tab 2: Electrical characteristics for different pencil hardnesses

5. Amplification circuit

As the resistivity signal is very weak to be measured, an amplification circuit is needed. We propose in figure 4 an example of the amplification circuit used for this datasheet. In our example proposed, the acquisition of this signal is acquired with an Arduino One. The potentiometer we used is MCP41010, which can be changed for values max to 10kΩ with 256 steps. The circuit consists on:

- Low-pass filter at the input (R1, C1) with a cutoff frequency of 16 Hz to remove the current noise in the input signal.
- Low-pass filter (R4, C2) with a cutoff frequency 1.6 Hz to eliminate the 50 Hz noise from the power grid.
- The output filter (R5, C3) with a cutoff frequency 1.6 kHz to filter the ADC noise

The formula used to calculate the resistance of the graphite strain sensor is as follows:

$$R_{sensor} = \left(1 + \frac{R4}{R2}\right) * R1 * \frac{VCC}{VADC} - R1 - R3$$

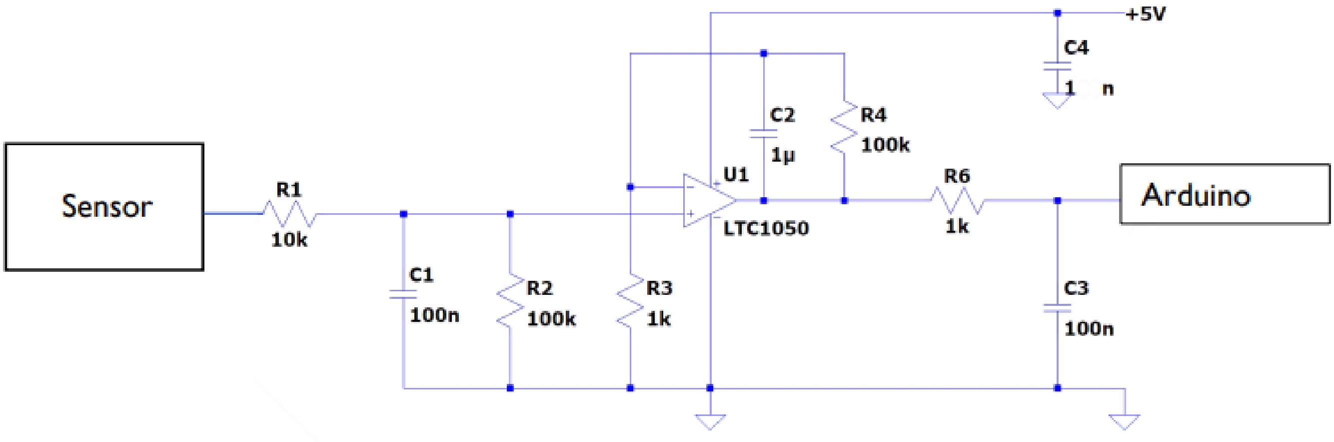

Figure 4: Proposed amplification circuit for measuring the signal

6. Sensor calibration and bench test

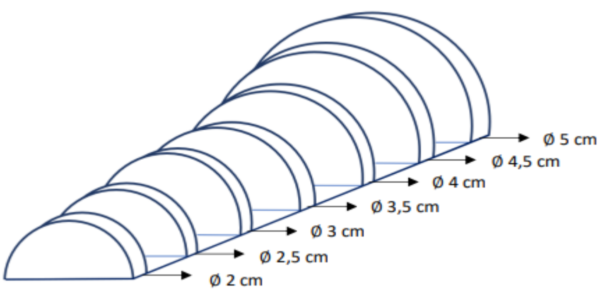

Figure 5: dimensions of the bench test

To calculate the relative variation of resistance with respect to the deformation of the sensor and thus characterize it, we use the following formula:

$$\varepsilon = \frac{e}{2R}$$

Where the thickness of the paper sheet, ε the deformation, and R the radius of curvature. (e = 0,16mm).

For better precision, we define R relative by Rr= Rmeas- Ro / Ro, where Ro is the resistance without deformation (initial state) and Rmeas the resistance for each deformation.

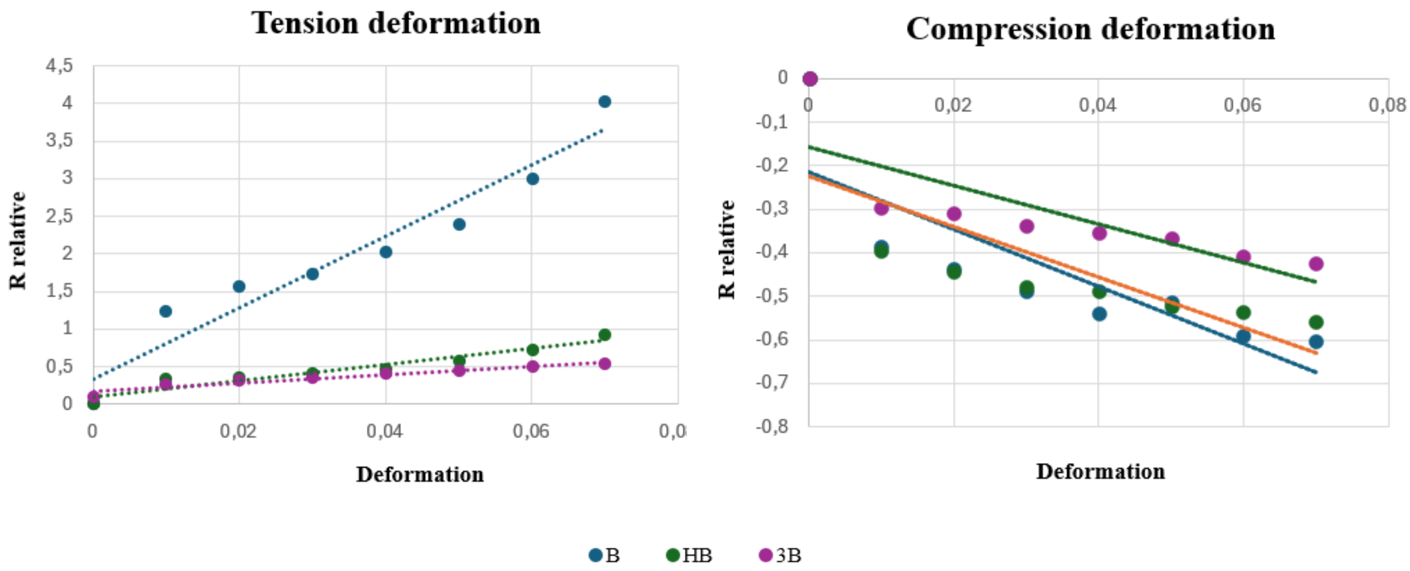

Figure 6,7: Characteristics graphs of resistance in standard test conditions

7. Sensor sensitivity

The sensitivity of the graphite sensor is directly determined by the slope of its characteristic curve (figure 6 and 7) . A steeper slope indicates higher sensitivity, as the sensor registers a much more significant change in resistance with minimal physical deformation.

Under tension, the B pencil exhibits the highest sensitivity by far, characterized by a sharply steep slope, making it highly responsive to minute stretching. Under compression, however, these slopes are very similar one to the other. Values can be found in tab 3.

Pencil hardness	Tension sensibility	Compression sensibility
HB	10	4,4
B	47,8	6,5
3B	5,7	5,7

Tab 3: Approximative sensitivity values: $\Delta R_{re\square a\square ive}$ / ΔDeformation

8. Notes

The main disadvantage of this sensor is the initial state. With every measure, graphite particles tend to decrease or leave the sensor. Thus, resistivity in the flat position Ro is never the same after one measure, which means the time life of the sensor is very short. We haven’t got a precise calculation for it, but after a few measures we were obligated to deposit more graphite, which completely changed the resistivity in the flat position. Moreover, if the change of the deformation is quite abrupt, the strain sensor is more likely to break.

9. Bibliography

Cheng-Wei Lin and al. (2014) - Pencil Drawn Strain Gauges and Chemiresistors on Paper